

Executive Summary

The following analysis is to help the hospital analyze operational efficiencies and identify areas for improvement. I cleaned the data to standardize dates, times and number columns, remove duplications, missing values, typos, irrelevant entries and different values stored in one column. In pivot tables, I analyzed appointments, billing, patients, insurance providers and treatments. The analysis concluded there is a bottleneck in processes for scheduling appointments and collecting payments negatively impacting hospital outcomes, so any operations involved in these areas should be reviewed and improved. Further recommendations to gather patient feedback to rate doctors, insurance providers and treatments to review success rates and standards of care.

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Assumptions and Change Log

Assumptions

All data is synthetically generated for educational and project use.
No real patient information was used for this analysis.

Change Log

Downloaded hospital management from Kaggle
Imported hospital management data to Excel

Raw Data: Appointments

	A	B	C	D	E	F	G	H
1	appointment_id	patient_id	doctor_id	appointment_date	appointment_time	reason_for_visit	status	
2	A001	P034	D009	8/9/2023	3:15:00 PM	Therapy	Scheduled	
3	A002	P032	D004	6/9/2023	2:30:00 PM	Therapy	No-show	
4	A003	P048	D004	6/28/2023	8:00:00 AM	Consultation	Cancelled	
5	A004	P025	D006	9/1/2023	9:15:00 AM	Consultation	Cancelled	
6	A005	P040	D003	7/6/2023	12:45:00 PM	Emergency	No-show	
7	A006	P045	D006	6/19/2023	4:15:00 PM	Checkup	Scheduled	
8	A007	P001	D007	4/9/2023	10:30:00 AM	Consultation	Scheduled	
9	A008	P016	D010	5/24/2023	8:45:00 AM	Consultation	Cancelled	
10	A009	P039	D010	3/5/2023	1:45:00 PM	Follow-up	Scheduled	
11	A010	P005	D003	1/13/2023	3:30:00 PM	Therapy	Completed	
12	A011	P022	D007	11/12/2023	4:00:00 PM	Checkup	No-show	
13	A012	P029	D003	5/7/2023	10:00:00 AM	Follow-up	Completed	
14	A013	P003	D002	8/16/2023	12:00:00 PM	Emergency	Scheduled	
15	A014	P012	D010	5/25/2023	10:30:00 AM	Emergency	Cancelled	
16	A015	P026	D004	1/15/2023	5:15:00 PM	Consultation	No-show	
17	A016	P016	D008	6/30/2023	11:00:00 AM	Consultation	Scheduled	
18	A017	P037	D009	7/11/2023	5:00:00 PM	Emergency	Scheduled	
19	A018	P022	D007	11/14/2023	9:45:00 AM	Consultation	Cancelled	
20	A019	P029	D001	2/6/2023	3:30:00 PM	Checkup	Cancelled	
21	A020	P014	D003	12/5/2023	3:15:00 PM	Consultation	Completed	
22	A021	P028	D009	4/24/2023	10:00:00 AM	Therapy	No-show	

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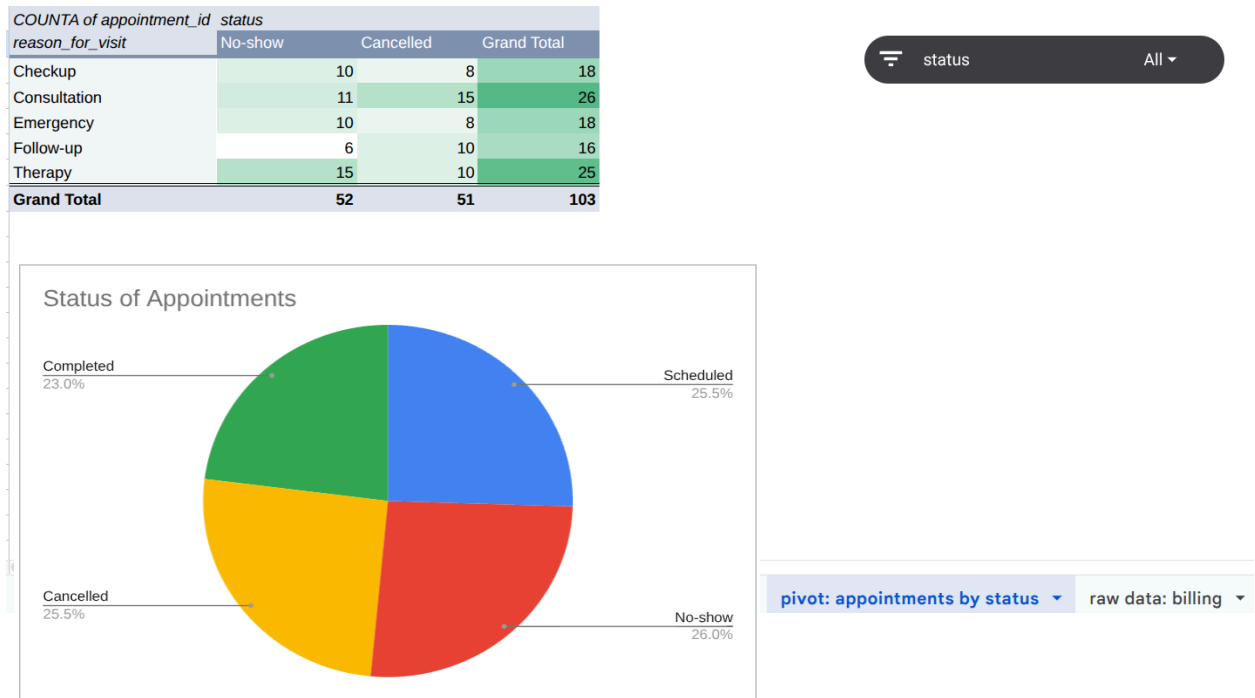
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assumptions & change log ▾

raw data: appointments ▾

pivot: appointments ▾

Pivot: Appointments by Status



Raw Data: Billing

bill_id	patient_id	treatment_id	bill_date	amount	payment_method	payment_status		
B001	P034	T001	8/9/2023	\$ 3,941.97	Insurance	Pending		
B002	P032	T002	6/9/2023	\$ 4,158.44	Insurance	Paid		
B003	P048	T003	6/28/2023	\$ 3,731.55	Insurance	Paid		
B004	P025	T004	9/1/2023	\$ 4,799.86	Insurance	Failed		
B005	P040	T005	7/6/2023	\$ 582.05	Credit Card	Pending		
B006	P045	T006	6/19/2023	\$ 1,381.00	Insurance	Pending		
B007	P001	T007	4/9/2023	\$ 534.03	Cash	Failed		
B008	P016	T008	5/24/2023	\$ 3,413.64	Cash	Failed		
B009	P039	T009	3/5/2023	\$ 4,541.14	Credit Card	Paid		
B010	P005	T010	1/13/2023	\$ 1,595.67	Cash	Paid		
B011	P022	T011	11/12/2023	\$ 4,671.66	Cash	Failed		
B012	P029	T012	5/7/2023	\$ 771.20	Insurance	Pending		
B013	P003	T013	8/16/2023	\$ 4,704.96	Cash	Paid		
B014	P012	T014	5/25/2023	\$ 2,082.30	Credit Card	Paid		
B015	P026	T015	1/15/2023	\$ 956.39	Insurance	Pending		
B016	P016	T016	6/30/2023	\$ 2,686.42	Insurance	Paid		
B017	P037	T017	7/11/2023	\$ 1,655.49	Credit Card	Pending		
B018	P022	T018	11/14/2023	\$ 1,781.93	Insurance	Pending		
B019	P029	T019	2/6/2023	\$ 1,882.80	Insurance	Pending		
B020	P014	T020	12/5/2023	\$ 4,113.62	Credit Card	Failed		
B021	P028	T021	4/24/2023	\$ 2,926.23	Insurance	Failed		

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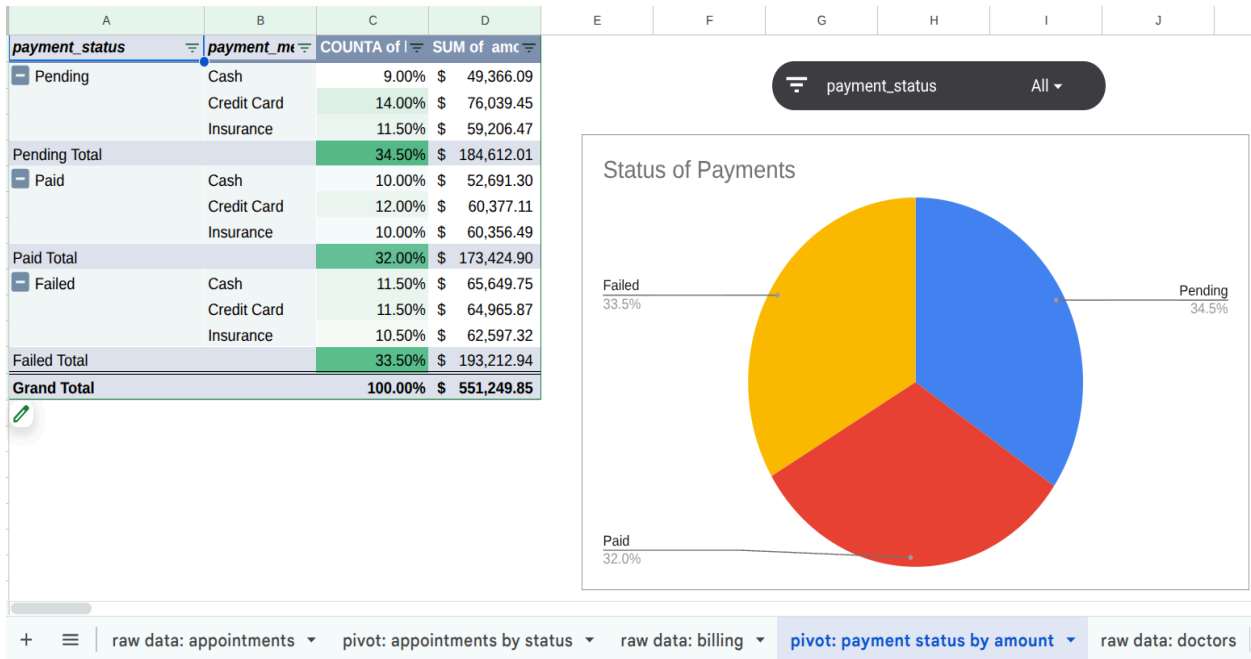
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assumptions & change log ▾

raw data: appointments ▾

pivot: appointments by status ▾

Pivot: Payment Status by Amount

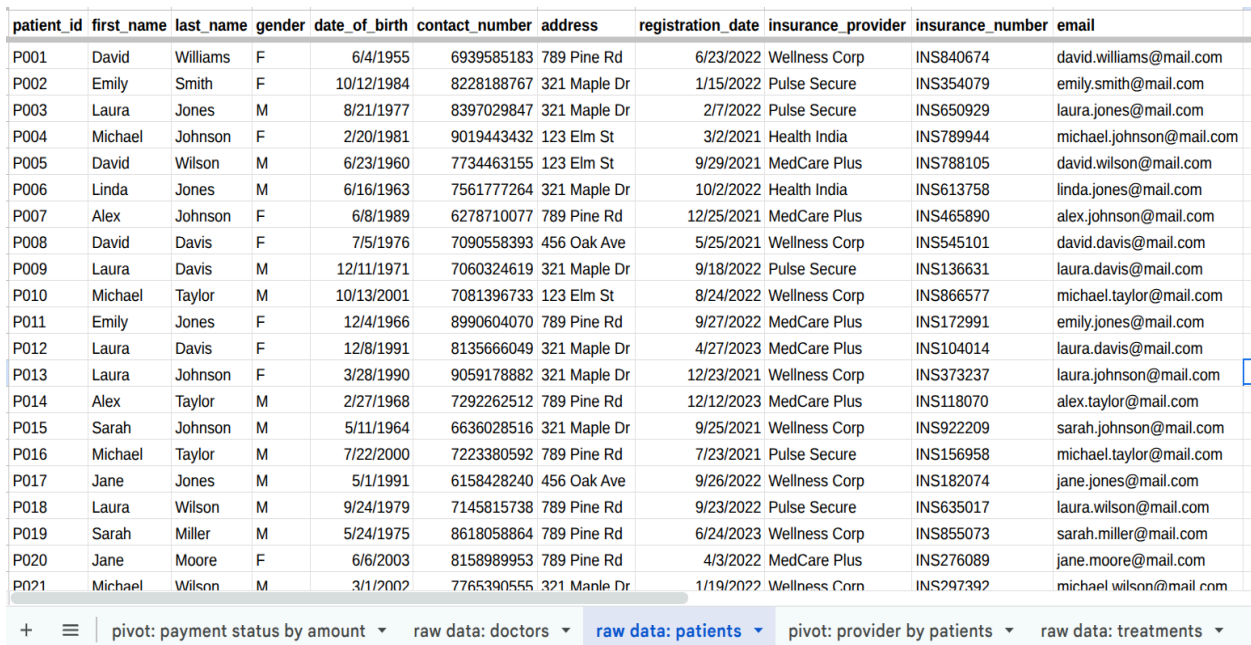


Hospital Management Analysis PDF

Raw Data: Doctors

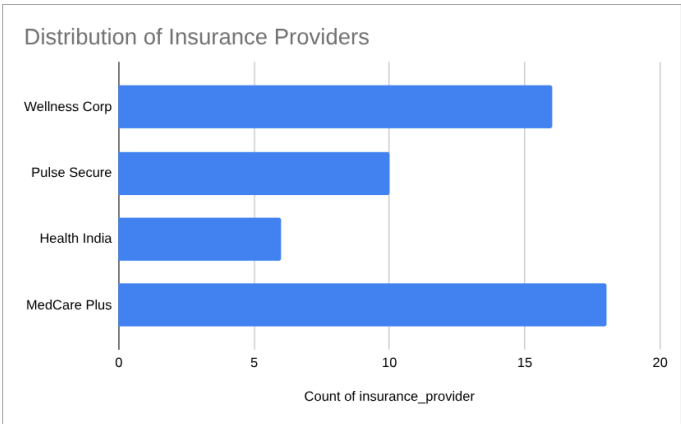


Raw Data: Patients



Pivot: Provider by Patients

insurance_provider	COUNTA of patient_id
MedCare Plus	18
Wellness Corp	16
Pulse Secure	10
Health India	6
Grand Total	50



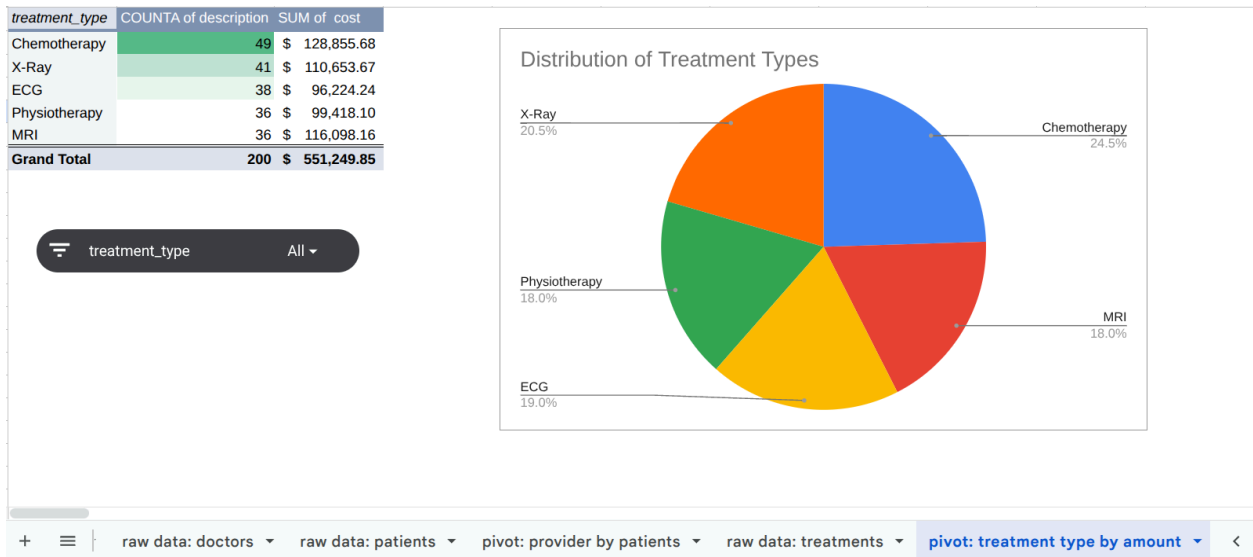
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Raw Data: Treatments

treatment_id	appointment_id	treatment_type	description	cost	treatment_date					
T001	A001	Chemotherapy	Basic screening	\$3,941.97	8/9/2023					
T002	A002	MRI	Advanced protocol	\$4,158.44	6/9/2023					
T003	A003	MRI	Standard procedure	\$3,731.55	6/28/2023					
T004	A004	MRI	Basic screening	\$4,799.86	9/1/2023					
T005	A005	ECG	Standard procedure	\$ 582.05	7/6/2023					
T006	A006	Chemotherapy	Standard procedure	\$1,381.00	6/19/2023					
T007	A007	Chemotherapy	Advanced protocol	\$ 534.03	4/9/2023					
T008	A008	Physiotherapy	Basic screening	\$3,413.64	5/24/2023					
T009	A009	Physiotherapy	Standard procedure	\$4,541.14	3/5/2023					
T010	A010	Physiotherapy	Standard procedure	\$1,595.67	1/13/2023					
T011	A011	MRI	Basic screening	\$4,671.66	11/12/2023					
T012	A012	Chemotherapy	Standard procedure	\$ 771.20	5/7/2023					
T013	A013	MRI	Standard procedure	\$4,704.96	8/16/2023					
T014	A014	ECG	Basic screening	\$2,082.30	5/25/2023					
T015	A015	Physiotherapy	Basic screening	\$ 956.39	1/15/2023					
T016	A016	MRI	Basic screening	\$2,686.42	6/30/2023					
T017	A017	MRI	Basic screening	\$1,655.49	7/11/2023					
T018	A018	ECG	Advanced protocol	\$1,781.93	11/14/2023					
T019	A019	X-Ray	Basic screening	\$1,882.80	2/6/2023					
T020	A020	Chemotherapy	Advanced protocol	\$4,113.62	12/5/2023					
T021	A021	X-Ray	Advanced protocol	\$2,926.23	4/24/2023					

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Pivot: Treatment Type by Amount



Recommendations

- Optimize Appointment Management:** Over 50% of scheduled appointments result in cancellations or no-shows. To reduce this rate, the hospital should evaluate and enhance appointment-related processes, including the member portal, administrative workflows, automated reminders, and online scheduling systems.
- Improve Billing and Payment Processes:** Analysis of billing data reveals that 68% of payments are either pending or have failed. A comprehensive review of the hospital's accounting and payment systems is recommended to identify and resolve inefficiencies or bottlenecks in the payment pipeline.
- Enhance Provider Performance Metrics:** Current data on doctors, treatments, and insurance providers primarily tracks operational metrics such as years of experience, provider-to-patient ratios, and service volume. Incorporating patient satisfaction surveys and outcome-based feedback will offer deeper insights into provider effectiveness and care quality.